

Sustainability-in-the-Loop for Artificial Intelligence: Integrating Sustainability into LLM Routing

Houssam Kanso ¹ and Salim Khazem ¹

Abstract: Large language models (LLMs) are increasingly deployed in enterprise workflows, yet the energy overhead of both model inference and routing decisions is rarely measured and is typically ignored by existing routers. Concerns are rising about the environmental footprint and energy consumption of these technologies. At the same time, sustainability regulations become stricter and organizations are under increasing pressure to better understand and communicate their environmental, social, and governance (ESG) impacts. In this paper, we present a sustainability-in-the-loop routing approach that makes energy a primary concern criteria in addition to the predicted task performance. We first benchmark four representative routing strategies and quantify their per-request energy consumption, selecting the most eco-efficient router for deployment. We then measure energy use across seven open-weight LLMs under identical hardware and workload conditions and derive per-token energy costs that can be used as routing priors. Based on these measurements, we integrate an energy-aware scoring function into the router to select, for each query, the model that best balances effectiveness and performance. Our results highlight substantial variability in routing overhead and per-token energy across models and show that energy-aware routing can reduce consumption while maintaining response quality.

Keywords: Artificial Intelligence, Large Language Models (LLMs), Sustainability, Multi-agent systems, Energy Aware, LLM Routing

1 Introduction

In recent years there has been an evolution in the global regulations and increasing sustainability requirements. For example, companies must produce detailed reports on their environmental, social, and governance (ESG) impacts. The carbon impact of companies is not only limited to the direct emissions associated with their core operations but also should include all the new used technologies such as artificial intelligence tools, in particular large language models (LLMs).

Faced with the accelerating pace of digital transformation and the increasing adoption of artificial intelligence (in particular LLMs), businesses find themselves at a crossroads where technological performance and environmental responsibility must converge. While AI and GenAI offer powerful levers for optimizing operations, fostering innovation, and supporting decision-making, their widespread deployment raises significant challenges regarding energy consumption, carbon footprint, and the use of critical resources such

¹ Talan Research and Innovation Center, Paris, France.
houssam.kanso@talant.com,  <https://orcid.org/0000-0002-4811-191X>;
salim.khazem@talant.com,  <https://orcid.org/0000-0001-5958-6120>

as water. In this context, the transition to sustainable AI that is more efficient and better integrated into ESG frameworks is no longer an option but a strategic necessity.

Thus, the development of AI tools capable of measuring, monitoring, and minimizing their own energy consumption appears to be an essential condition for aligning technological performance with respect for environmental limits. The design of such solutions not only helps companies respect their sustainability engagement but also can reduce the costs while maintaining the performance.

This paper introduces sustainability awareness into LLM routing mechanisms designed to select among multiple available language models based on both performance and energy consumption. It further distinguishes itself through an eco-efficient approach by exploring the energy consumption of different routers and models and evaluating the environmental impact of various proposed solutions in order to select the best models that achieve an effective trade-off between model performance and energy efficiency.

The contributions of this work can be summarized as follows. First, we benchmark four widely used LLM routers and quantify the energy overhead of the routing step itself, allowing us to select an eco-efficient router as the orchestration component. Second, we propose a reproducible measurement protocol to estimate the inference energy consumption of LLMs in a controlled environment, and we report energy-per-token measurements (Wh per 1K tokens) for seven open-source models. Finally, we integrate these empirically measured energy costs into the routing decision through an explicit performance-energy trade-off, and we track the exact energy consumed by the selected model at inference time, enabling sustainability-in-the-loop model selection.

Our paper is organized as follows. Section 2 discusses related work. In Section 3 we present our energy-aware router architecture for choosing the best trade-off between performance and energy consumption. In Section 4, we describe our experimental setup and detail the implementation of our architecture for sustainable LLM routing by first choosing the most efficient router, then measuring the energy consumption per token for a series of models and implementing the energy awareness to the router. In Section 5, we discuss the results of the experimentation. Finally, we conclude and layout future directions and perspectives in Section 6.

2 Related Work

In recent years, sustainable artificial intelligence has emerged as an important research direction, driven by the rapid growth of large-scale models and the increasing awareness of their energy consumption and environmental impact. As AI systems become more deeply embedded in organizational processes and companies, prior work has explored methods for measuring the energy impact of these technologies, improving energy efficiency, reducing carbon footprints, and integrating sustainability considerations into model design,

deployment, and decision-making. In the following section, we review the main state-of-the-art research work by dividing them into 4 categories.

2.1 Sustainable AI and Green AI

The rapid growth of artificial intelligence systems has raised increasing concerns about their energy consumption and environmental impact. In response, a growing body of research has emerged under the umbrella of sustainable or green AI, aiming to reduce the carbon footprint of AI models while maintaining acceptable performance. This line of work explores both methodological and system-level approaches to improve the energy efficiency of AI throughout its life-cycle.

Recent approaches focus on the different stages of LLM training and inference. These solutions aim to reduce energy consumption and carbon footprint through techniques such as optimizing training and fine-tuning costs, applying prompt engineering with sustainability-oriented directives, and assessing the environmental impact of AI to create new regulations [Ha24].

In [Pa21], the authors examined the energy use and carbon footprint specifically during the training of large ML models. They found that that training of AI models can be made far more efficient by using sparsely activated models, choosing low-carbon geographic locations and times, and relying on cloud data centers.

SPROUT [Li24] is a framework that introduces "generation directives" into the prompt to guide the generation process in order to reduce the carbon footprint of inference in LLMs.

In [LY24], the authors explored the carbon footprint generated during fine-tuning of large language models such as BERT, DistilBERT, and T5. The authors measure CO₂ emissions using CodeCarbon[Co24] while comparing different tokenizers and GPUs. The results show that DistilBERT and the A100 GPU can massively reduce emissions while maintaining good performance.

The environmental impact of AI was discussed in [LTM]. The focus was on the significant environmental impacts that can be made throughout the entire life cycle of AI (from the extraction of raw materials and manufacturing of infrastructure to energy-intensive training, deployment, and everyday use—each consuming large amounts of energy, water, and minerals and producing greenhouse gas emissions). It highlights that current research often underestimates all these effects and focus mainly on training emissions. Therefore, there is a need for a broader assessment of the full AI life cycle and usage patterns to understand environmental costs.

2.2 Energy and Carbon Measurement in AI Systems

Accurate measurement of energy consumption and carbon emissions is a prerequisite for the development of sustainable AI systems. Prior work has investigated various techniques for estimating and monitoring the energy usage of machine learning models during training and inference, as well as for translating these measurements into carbon impact metrics. These approaches provide the foundation for evaluating and comparing the environmental cost of different AI architectures and deployment strategies.

In [Sa23], the authors focused on measuring the inference costs that often exceed training costs in the long term. They proposed an equation to calculate the energy consumption of the inference of the LLM models based on the number of input and output tokens. Their equation is strongly influenced by the environment where the models are deployed. Therefore, in our paper we made all our experimentation in the same environment.

In [Je25], the authors analyzed 30 major language models and estimated the energy, water, and CO₂ consumption of each model. These models were deployed in different environments therefore each was impacted by the environment where it was deployed. They also show that even with minimal consumption per query, when scaled to 700 million queries per day, the aggregates result in a high annual electricity cost.

A study that analyses the carbon footprint of machine learning (ML) models hosted on the Hugging Face platform was presented in [Ca23]. However, no clear correlation was found between performance and carbon footprint, therefore it is important to differentiate between performance, monetary cost and energy cost.

In [Hu24], the authors proposed a monitoring framework designed to measure energy consumption during inference. Their analysis reveals a variability in energy consumption across different model sizes and deployment environments. They found that larger models consume up to two times more energy per token compared to smaller models. Additionally, the study showed that inference energy consumption correlates strongly with response characteristics such as the number of tokens and generation time, while prompt complexity has only a minimal effect.

Google [El25] announced that they reduced per-prompt energy use and carbon emissions by 33 and 44, respectively, over one year. The results were obtained using internal production data, which limits reproducibility and comparability with open source deployments.

2.3 LLM Routing and Model Selection Strategies

With the increasing availability of large language models of varying sizes and capabilities, recent research has focused on routing and model selection mechanisms that dynamically choose the most appropriate model for a given task. Existing approaches primarily optimize

Tab. 1: Comparison of Routing Approaches

Attribute	RouteLLM	ArchRouting	Not Diamond	IRT-Router
License	Open source	Open source	Proprietary	Open source
Type of router	Complexity classifier	Heuristic routing	Ranked recommendation system	Ranked recommendation system
Parameters	Threshold	No threshold; configurable via prompts	No threshold (cost trade-off only)	Model and cost thresholds (attention-based)
Retrainable	Yes	No	Yes	Yes

for performance-related criteria such as accuracy and latency, others focus on monetary cost. However, the integration of energy and environmental considerations into LLM routing decisions remains an emerging research direction.

A platform for comparing LLM routers, called RouterArena, was introduced in [Lu25]. It provides a systematic evaluation of routing approaches based on both model performance and inference financial cost per token.

Several routing frameworks have been proposed in the literature. In this paper, we chose 4 well known routers and compare them and experiment the energy consumption of each of them. In particular, we evaluate their ability to make routing decisions that jointly consider model performance and energy efficiency. In Table 1, we compare four of the most commonly used LLM routers based on several criteria.

RouteLLM [On25] enables routing between at most two models, making it simple to implement and configure on pretrained models. One model is considered weak for simple requests and another is strong for more complex requests. However, extending the system to support additional models requires retraining, which makes scaling to more complex routing scenarios relatively difficult.

Not Diamond [No] provides an easy-to-use API-based routing solution that is straightforward to integrate. However, the energy consumption of the routing process itself is opaque, making it unsuitable for energy-aware evaluations.

Arch-Router [Tr25a] is easy to implement and configure, as routing decisions are guided by an LLM using route descriptions and configurations. While this design offers high flexibility when introducing new routes, it comes with a significant overhead due to the additional LLM inference required for routing.

Finally, IRT-Router [So25] relies on a pretrained supervised router that estimates a score based on model performance and monetary cost. While the framework does not account for

energy consumption, incorporating sustainability-related metrics is feasible. Adding new models, however, remains relatively complex.

2.4 Multi-agent Architectures

Multi-agent systems has been widely studied as effective paradigms for managing complex decision-making processes and coordinating heterogeneous tasks. Recent advances have extended these systems to support adaptive reasoning, orchestration, and context-aware decision support in organizational settings. Nevertheless, sustainability considerations are still rarely treated as first-class constraints within agent-based AI architectures.

In [Kr25], the authors trace the evolution of AI agents, from reactive systems to complex agent architectures integrating memory, planning, and interaction with tools. The article highlights the potential of agents for real-world applications in industrial and organizational contexts.

In [Tr25b], the authors analyze collaboration mechanisms between agents based on LLMs, demonstrating that specialization and structured communication allow them to outperform single-agent approaches on complex tasks. The study highlights the importance of collaboration strategies for improving the quality of results.

Recent state-of-the-art research explores several complementary approaches, including dynamic generation of specialized agents (such as AutoAgents[Ch24]). With the increasing use of multiple agents there are growing questions about the energy consumption and the environmental impact of this emerging technology.

2.5 Limitations and Open Challenges

Despite significant progress in sustainable AI, several limitations and open challenges remain.

First, existing approaches to energy and carbon measurement often rely on heterogeneous environments, proprietary telemetry, or narrow measurement boundaries, which limits reproducibility and comparability across studies. While recent production-scale analyses demonstrate substantial efficiency gains, their dependence on highly optimized, provider-specific infrastructures makes it difficult to generalize findings to open-source or academic deployments.

Second, current LLM routing and model selection approaches mainly optimize for performance, latency, or monetary cost, and largely ignore energy consumption as a decision criterion. When energy is considered, it is often implicit or difficult to measure due to opaque routing mechanisms or unreported overheads. As a result, there is a lack of practical

methods that jointly consider performance, cost, and energy when selecting or routing between models. Addressing these challenges requires transparent energy measurements and routing strategies that explicitly integrate energy consumption, enabling more sustainable deployment of LLM-based systems.

3 Architecture

To address the challenge of building sustainable artificial intelligence systems, we propose to study the energy impact of different routers and models. Our architecture integrates orchestration through a router that choose the highest-performing and most sustainable model to use for each user query. Figure 1 illustrates the complete architecture of the solution. In the following section, we detail all the components of the architecture.

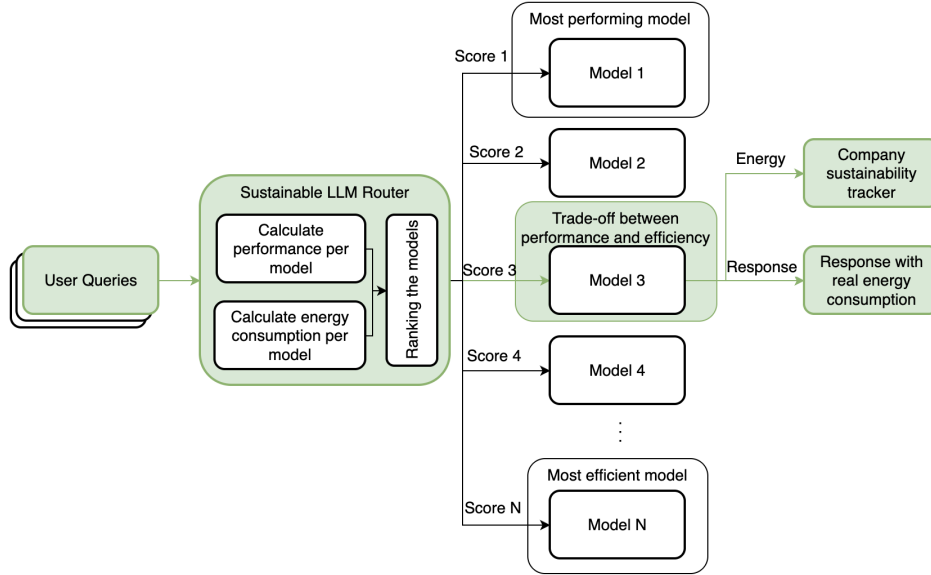


Fig. 1: Sustainable LLM Router Workflow

3.1 User Queries

The entire workflow is triggered by the user who submits a query. The process begins with a prompt entering the system that represents the initial request sent by the user. It can range from a simple question to a complex task. Based on the complexity and the topic of each query it will be routed to the corresponding model.

3.2 Sustainable LLM Router

The LLM router coordinates the system's behavior and ensures query processing. Its main responsibilities are:

- *Calculate performance per model:* The router selects the most relevant AI models or algorithms for the task. These models vary based on the complexity and the purpose of the query. It may differ if the request involves natural language processing, reasoning, summarization, coding, mathematical calculation or translation. The performance term is the ability score estimated by IRT-Router's built-in function.
- *Calculate energy consumption per model:* The router calculates the energy consumption required by each model to respond to the user request based on a created database that includes energy consumption per token for each model. It also integrates feedback from the selected model where it captures energy consumption and environmental impact metrics.
- *Ranking the models:* The models are ranked based on the calculated performance and predicted energy consumption. In the example of Figure 1, scores ranging from *score 1* to *score N* are calculated with the highest score being *Score 3* that represents the best trade-off between performance and energy efficiency.

The router also has the role to define the parameters that guide model behavior. This includes tuning thresholds, setting constraints, and adjusting operational settings to balance accuracy and efficiency. However, these parameters were fixed in the experiments conducted in this paper.

3.3 LLM models and execution environment

Within the *LLM models and execution environment* component, there are a set of models that have been ranked by the router in the previous step, only the model with the highest score will be called.

- *Model is initiated* according to the defined parameters.
- *Query is executed* in an environment in which the chosen model executes the user query. It provides the operational context in which models perform their tasks and provide a response to the user. During the execution of the query, the energy consumption is measured using CodeCarbon [Co24].

3.4 Response

User response: The user obtains the response from the system. This response is impacted by the results of router choice (based on performance and energy consumption), the model

itself, and the model parameters. It also includes the measured energy consumption of the selected model, which gives the user visibility on the energy required to answer their query.

3.5 Sustainability tracker

Sustainability tracker: is a tracking system that allows the company to track and monitor energy consumption related to LLM usage and quantify the energy that was saved by using the sustainable LLM router. It plays a significant role in companies' annual carbon-footprint reporting.

This architecture provides an end-to-end workflow in which performance estimation and sustainability constraints are combined at routing time and the selected model is executed with energy tracking enabled. By explicitly separating routing, execution, and monitoring, the system can be extended with additional models, updated energy and performance indicators, or new sustainability indicators without changing the overall pipeline.

4 Implementation

The implementation of the proposed architecture followed 3 main steps: (1) Benchmark of different routers in order to use the router with the lowest energy consumption per routing, (2) measure the energy consumption of each of the LLM models and calculate the energy consumption per token, and (3) implement the energy constraint to the router in addition to the performance.

4.1 Experimental Setup

All experiments were conducted on a workstation running Debian GNU/Linux (trixie/sid) with Linux kernel version 6.6.9. The system is equipped with an AMD Ryzen 9 7950X CPU (16 cores, 32 threads, up to 5.88 GHz) and 32 GB of RAM. GPU acceleration was provided by a single NVIDIA GeForce RTX 4090 with 24 GB of VRAM. The system uses NVIDIA driver version 555.42.02 with CUDA support (CUDA runtime version 12.5, CUDA compiler version 12.0). All experiments were implemented and executed in Python 3.11.7. The system includes a 1 TB NVMe SSD, and all code was compiled using GCC version 13.2.0.

4.2 Dataset

A dataset of 30 prompts was created. It ranges from simple prompts to more complex ones (with mathematical calculations and problem solving reasoning). To evaluate, compare, and

measure the power consumption of each of the LLM models per token. The following are examples of prompts used in the dataset:

- You are an SQL specialist. Write a query to list the five customers with the highest aggregated purchase totals.
- Solve: What is the derivative of $x^2 \cdot \ln(x)$? Show the steps.
- You are a machine learning expert. Explain the difference between supervised and unsupervised learning.

Table 2 presents the language models used in this study, including DeepSeek-Coder, Qwen2.5-7B, Qwen2.5-Math-7.6B, LLaMA 3.1-8B, LLaMA 3.1-70B, Mixtral 8×7B, and Mistral-7B-Instruct. The table compares their licenses, total and active parameter counts, and primary use cases, covering a spectrum from lightweight instruction-following and code generation models to large-scale architectures optimized for advanced reasoning and efficient inference.

4.3 Benchmark of different routers

In order to start the implementations, we run a benchmark in order to compare the different energy consumption needed for routing a query. In order to evaluate the different routing models, we used a set of 6 prompts. The power consumption of each of the following router were measured: RouteLLM, Archrouting, Not Diamond, and IRT-Router.

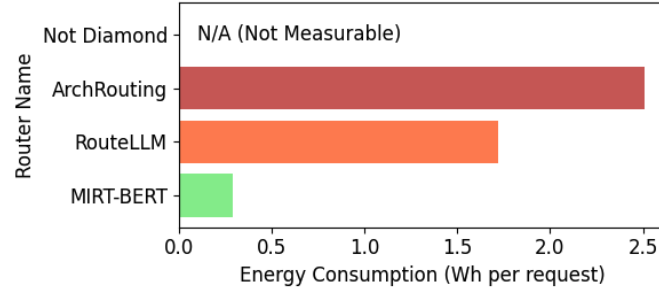


Fig. 2: Energy Consumption of Routers for 6 requests

Router RouteLLM consumes 1.720 Wh per request on average. For the Router Not Diamond it was not possible to measure the energy consumption because the used method is based on calling an external API that makes the routing. IRT-Router shows comparatively lower energy usage at 0.294 Wh, indicating a more energy-efficient routing behavior. In contrast, Router ArchRouting incurs the highest energy consumption, reaching 2.507 Wh per request, which can be attributed to additional routing overhead caused by the LLM technique used to route (cf. Fig. 2).

The results of our experimentation shows that the power consumption of IRT-Router is the lowest, therefore we decided to use it to do the routing between the models in order for our solution to be the most eco-efficient.

4.4 Power consumption per model

In order to measure the energy consumption per 1000 tokens, we automated the benchmarking using the algorithm 1 was developed, it ensures using the dataset described in 4.2 and the CodeCarbon measurement tool [Co24]. Our experiments consists of running a set of prompts on each of the models. We ran each of the prompts 3 times and compared the difference between each of the runs in order to ensure the reproducibility of the experimentation. We also used a function that forces all the common sources of randomness in Python ML code to behave the same way every time the program is executed.

Algorithm 1 Energy Benchmarking of LLMs using Ollama

Require: Ollama endpoint U , models $Models$, prompts $Prompts$

Ensure: Per-run measurements and per-model energy summary

```

1: Initialize empty list  $\mathcal{R}$ 
2: for all  $m \in Models$  do
3:   for all  $p \in Prompts$  do
4:     for  $k = 1$  to  $3$  do                                      $\triangleright$  Repeat measurements
5:       Sleep( $\delta t$ )
6:       energy_tracker.start()
7:        $t_0 \leftarrow \text{now}()$ 
8:       response  $\leftarrow \text{GENERATE}(m, p)$ 
9:        $t_1 \leftarrow \text{now}()$ 
10:      energy_tracker.stop()
11:       $E \leftarrow \text{energy\_tracker.energy}$ 
12:       $n_{\text{tok}} \leftarrow \text{response.tokens}$ 
13:      Append  $(m, p, k, n_{\text{tok}}, t_1 - t_0, E)$  to  $\mathcal{R}$ 
14:    end for
15:  end for
16: end for
17: Aggregate  $\mathcal{R}$  by model to obtain  $(N_{\text{tok}}, E, T)$ 
18: Compute WhPer1k  $\leftarrow 1000 \cdot E / N_{\text{tok}}$ 
19: Save  $\mathcal{R}$  and summary results to CSV
20: return  $(\mathcal{R}, \text{summary})$ 

```

Tab. 2: Comparison of used Language Models

Model	License	Parameters	Active Parameters	Primary Use
DeepSeek-Coder	Open source	6.7B	6.7B	Code generation
Qwen2.5-7B	Open source	7B	7B	Instruction following
Qwen2.5-Math-7.6B (Q6)	Open source	7.6B	~7.6B	Mathematical reasoning
LLaMA 3.1-8B	Open source	8B	8B	General-purpose chat
LLaMA 3.1-70B	Open source	70B	70B	Advanced reasoning
Mixtral 8x7B	Open source	8 × 7B (MoE)	~13B	Efficient large-model inference
Mistral-7B-Instruct	Open source	7B	7B	Lightweight instruction-following

4.5 Implementing energy awareness to the router

After measuring the energy consumption per model, we included these data into the routing process. The router (IRT-Router) was modified in order to calculate the score based on the performance and the energy cost instead of the financial cost, as shown in Algorithm 2.

Eq. 1 defines the overall score is defined as a weighted combination of performance and energy consumption.

$$\text{Score} = a \times \text{Performance} - b \times \text{Cost}_{\text{energy}} \quad (1)$$

where Performance represents the predicted LLM performance based on the pre-trained router and $\text{Cost}_{\text{energy}}$ corresponds to the energy usage. The energy costs are defined per 1K tokens for each model. The coefficients a and b are non-negative weighting factors that reflect the relative importance of performance maximization versus energy cost ($b = (1 - a)$).

Once the scores are calculated, the models are ranked by score and the model with the highest score is chosen to be the best model. The best model is then called to execute the query and the response is sent back to the user. The energy consumption that was necessary to generate the response is measured using CodeCarbon [Co24].

Algorithm 2 Energy-Aware LLM Routing (Performance–Energy Trade-off) with Energy Tracking

Require: Query q , candidate models $\mathcal{M}$, weight $a \in [0, 1]$

Require: Energy cost per 1k tokens $\{c_m\}_{m \in \mathcal{M}}$ $\triangleright c_m$ is the energy cost (Wh/1k tokens)

Ensure: Best model m^* , scores $\{S(m)\}$, measured energy E , latency $t_1 - t_0$

```
1: Initialize empty list  $\mathcal{R}$ 
2: for all  $m \in \mathcal{M}$  do
3:    $\hat{P}(m, q) \leftarrow \text{PREDICT\_PERFORMANCE}(m, q)$ 
4:    $C(m) \leftarrow c_m$   $\triangleright$  Energy-cost proxy for model  $m$ 
5:    $S(m) \leftarrow a \cdot \hat{P}(m, q) - (1 - a) \cdot C(m)$ 
6:   Append  $(m, S(m))$  to  $\mathcal{R}$ 
7: end for
8: Rank  $\mathcal{R}$  by decreasing score
9:  $m^* \leftarrow \max_{m \in \mathcal{M}}(S(m))$ 
10: energy_tracker.start()
11:  $t_0 \leftarrow \text{now}()$ 
12: response  $\leftarrow \text{GENERATE}(m^*, q)$ 
13:  $t_1 \leftarrow \text{now}()$ 
14: energy_tracker.stop()
15:  $E \leftarrow \text{energy\_tracker.energy}$   $\triangleright$  Measured energy for executing the selected model (inference)
16: return  $(m^*, \{S(m)\}, E, t_1 - t_0)$ 
```

5 Results

In this section we present and discuss the results of the power consumption measurements per model and the implementation of energy awareness to the IRT-Router.

5.1 Power consumption per model

Figure 3 shows the energy consumption of the seven studied models per 1K generated tokens under identical experimental conditions.

The results reveal substantial differences in energy efficiency across models, particularly between large-scale and smaller models. The largest model, *llama3.1:70b*, exhibits the highest energy consumption at 10.31 Wh per 1K tokens. Among the mixture-of-experts and mid-sized models, *mixtral:8x7b* and *Qwen-2.5-Math-7.6B* consume 1.22 Wh and 0.88 Wh per 1K tokens, respectively. In contrast, *deepseek-coder* demonstrates the lowest energy consumption at 0.21 Wh per 1K tokens, indicating superior energy efficiency. The remaining models, *mistral:7b-instruct*, *llama3.1:8b*, and *qwen2.5:7b-instruct*, consume 0.67 Wh, 0.72 Wh, and 0.73 Wh per 1K tokens, respectively. Overall, these results highlight that energy consumption per token varies significantly across LLMs, even when evaluated

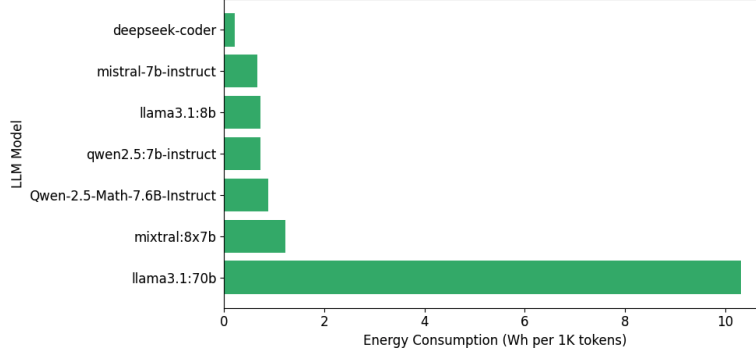


Fig. 3: LLM energy consumption per 1k tokens

on identical hardware and workloads, underscoring the importance of considering energy efficiency alongside performance when selecting or routing between models.

5.2 Implementing energy awareness to the router

Figure 4 shows the energy consumption of executing the same set of 30 prompts defined in 4.2 in 10 different routing scenarios. We explored different values of $a \in [0, 1]$ with an increase of 0, 1 each time. When $a = 0$, only the energy consumption incurred during routing is considered, whereas when $a = 1$, the metric accounts exclusively for routing performance.

Overall, the results demonstrate a significant reduction in energy consumption that can be divided by 10 when energy-aware routing is enabled. Compared to performance-only selection, incorporating energy consumption into the routing decision consistently lowers the total energy required to execute a prompt, while preserving competitive performance levels. However, since lower energy consumption may come at the cost of reduced model performance, the choice of routing strategy should be left to the user, allowing them to explicitly balance energy efficiency against performance according to their application requirements. This highlights the effectiveness of jointly optimizing performance and energy cost for sustainable LLM routing and deployment.

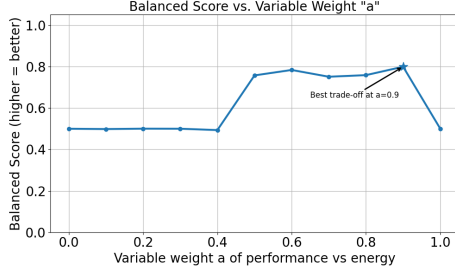


Fig. 4: Energy consumption across different routing scenarios for varying values of a .

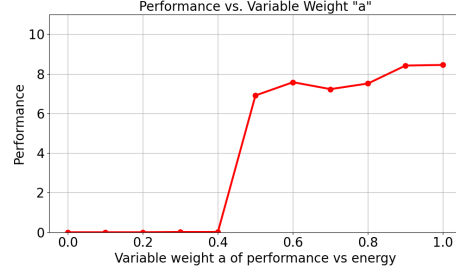


Fig. 5: Performance according to different routing scenarios for different values of a .

Figure 5 illustrates the evolution of system performance as a function of the weighting parameter 'a', highlighting the trade-off between performance and energy consumption.

The balanced score presented in Figure 6 is calculated to identify the best compromise between performance (to be maximized) and energy consumption (to be minimized). To make these two quantities comparable, they are first normalized over the interval $[0,1]$: performance is normalized in the conventional way (0 corresponding to the lowest achieved performance and 1 to the highest), while energy consumption is also normalized and then inverted so that lower energy values correspond to a higher score. The balanced score is then obtained as a weighted linear combination of these two normalized terms, with equal weights, reflecting an equivalent importance given to performance and energy efficiency. As illustrated in Figure 3, this approach highlights $a = 0.9$ as the optimal operating point, as it offers high performance while resulting in a disproportionate increase in energy consumption for $a = 1.0$.

Formally, let E and P denote the raw energy consumption and performance metrics, respectively. The normalization is expressed as $E_{\text{norm}} = (E - E_{\text{min}})/(E_{\text{max}} - E_{\text{min}})$ and $P_{\text{norm}} = (P - P_{\text{min}})/(P_{\text{max}} - P_{\text{min}})$. The balanced score is then computed as:

$$S_{\text{balanced}} = w_{\text{perf}} \cdot P_{\text{norm}} + w_{\text{energy}} \cdot (1 - E_{\text{norm}}), \quad (2)$$

where the inversion term $(1 - E_{\text{norm}})$ ensures that lower energy consumption contributes positively to the overall score, and $w_{\text{perf}} = w_{\text{energy}} = 0.5$ in our equal-weighting scheme. As illustrated in Figure 3, this approach highlights $a = 0.9$ as the optimal operating point, as it offers high performance while $a = 1.0$ results in a disproportionate increase in energy consumption for only marginal performance gains.

However, since reduced energy consumption can come at the expense of model performance, the choice of routing strategy should be left to the user, allowing them to balance energy efficiency and performance according to their application requirements. This underscores the effectiveness of jointly optimizing performance and energy cost for sustainable routing and LLM deployment.

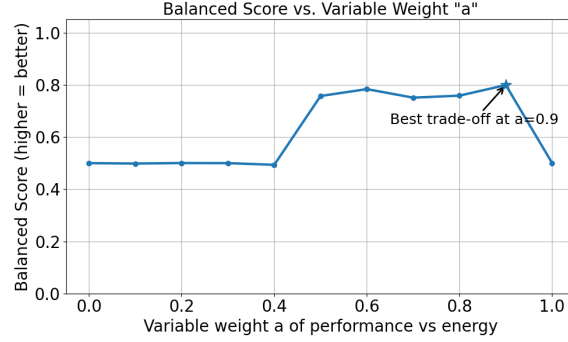


Fig. 6: Analysis of the trade-off between performance and energy consumption

6 Conclusion and Future Directions

In this paper, we presented a benchmark of four different LLM routers while focusing on the energy consumption of each of them per routing action. We also measured the energy consumption of 7 different models by experimenting the automated measurement process that can be extensible to any number of models and that provides the energy consumption per token for each of the models by running a set of 30 different prompts for each of these model. Finally, integrated the measured values to the router in order to take the energy consumption into consideration while accomplishing the choice between models. Some experiments on the impact of the weights of each of the performance and energy consumption were also conducted. Results show that including the energy consumption into the routing process reduces the final energy consumption needed by the LLMs to respond to user queries.

Future work will focus on extending the proposed framework to support multi-agent architecture with more than one model. We plan to investigate the energy consumption of different agentic design patterns. This will allow that we test our approach not only on context-free prompts but also on realistic enterprise workloads involving longer contexts (e.g., RAG pipelines, document-grounded queries). The framework will also be expanded to incorporate additional sustainability metrics, such as CO₂ emissions and water consumption, providing a more comprehensive assessment of environmental impact. As part of future work, we plan also to systematically investigate the optimal value of the trade-off parameter a that achieves a desirable balance between energy efficiency and performance. This performance score was not validated against independent quality judgments (we used the internal router performance score), future work will also validate it against human or benchmark-based evaluations. Finally, future experiments will be conducted using more recent models, as well as by exploring the influence of different model hyperparameters (e.g., temperature Top-k, Top-p, and sampling strategies) on performance, efficiency, and resource usage. It should be also used to test models that use the same techniques (in particular for the ones using CPU/system-memory offloading compared to GPU memory models).

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